# Basic Mathematics: A Student Refresher

Introductory study guide · Prepared as sample educational content for the StudentHub frontend project.

## 1. Number sense

Real numbers include rational numbers (which can be written as a fraction) and irrational numbers (such as √2). Estimate before calculating so that an answer can be checked for reasonableness.

## 2. Fractions, decimals, and percentages

A fraction represents part of a whole. To convert a fraction to a decimal, divide the numerator by the denominator. A percentage is a value per hundred: 0.25 = 25% = 1/4.

## 3. Algebraic expressions

A variable represents a quantity that may change. In 3x + 5, the coefficient of x is 3 and the constant is 5. Like terms can be combined, but unlike terms cannot.

## 4. Solving a linear equation

An equation states that two expressions are equal. Apply the same operation to both sides to keep equality. For example, 2x + 4 = 14 gives 2x = 10 and x = 5. Substitute the result to check it.

## 5. Functions and graphs

A function maps each allowed input to exactly one output. In y = 2x + 1, the slope is 2 and the vertical intercept is 1. A graph helps reveal patterns and relationships.

## 6. A problem-solving routine

Read the question carefully, identify known and unknown quantities, choose a representation, solve one step at a time, and check units and reasonableness. Showing working makes errors easier to find.

## Practice

Convert 3/5 to a percentage. Solve 4x – 7 = 13. If a notebook costs 800 units and its price rises by 10%, what is the new price?